

The Webster Method: Arithmetic Mean Rounding and Its Historical Role in U.S. Apportionment

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May 8, 2026

Abstract

The Webster (major fractions) apportionment method rounds each state’s exact proportional quota to the nearest integer using the arithmetic mean threshold of $s + 0.5$: a state receives $s + 1$ seats if its population-to-divisor ratio exceeds $s + 0.5$ seats, and s seats otherwise. Webster is the principal academic benchmark for evaluating Huntington-Hill’s small-state bias, and it was used in U.S. congressional apportionment in 1842 and from 1911 through 1940. In proportional representation systems for party-list elections, Webster’s arithmetic mean rounding is known as the Sainte-Laguë method, the dominant PR formula in Scandinavia.

We prove that Webster minimises the sum of squared deviations $\sum_i (s_i - q_i)^2$ from the exact quotas — an optimality property that makes it the divisor-method solution closest to strict proportionality in the least-squares sense. We show that Webster is paradox-immune as a divisor method and that it is neutral between small and large states in the sense that its arithmetic mean threshold neither systematically favors small populations (as Huntington-Hill’s geometric mean does slightly) nor large populations (as Jefferson’s floor does strongly). Using 2020 Census data, Webster produces seat allocations identical to Huntington-Hill for all 50 states; we identify the state nearest the 0.5 rounding boundary and quantify the population gap separating the two methods’ outcomes in 2020. We document Webster’s use in the 1842 Apportionment Act (the first to mandate single-member districts) and its use under the 1911 Apportionment Act, and explain why Congress replaced it with Huntington-Hill in 1941.

Keywords

Webster method, major fractions, Sainte-Laguë, arithmetic mean rounding, congressional apportionment, U.S. apportionment history, BISECT-APPORTION

1 Introduction: Webster in Historical Context

Daniel Webster proposed his major fractions method in 1832 in response to the perceived inequities of Jefferson’s floor-rounding method, which he argued gave disproportionate advantage to large states. Webster’s insight was that the fair rounding point for a fractional quota $q \in (s, s + 1)$ is the midpoint $s + 0.5$: states with fractional parts above 0.5 should round up, and those below should round down. This is precisely the standard arithmetic rounding rule taught in every elementary school — and its simplicity has made it a persistent alternative to Huntington-Hill in the academic literature.

Webster’s method was used for the U.S. apportionment of 1842, the first apportionment conducted under a statute that required single-member congressional districts [United States Congress, 1842]. It was reinstated in the Apportionment Act of 1911 and used for the 1910, 1920 (though the 1920 apportionment was never enacted), and planned 1930 apportionments. The failure of Congress to pass a reapportionment act after the 1920 Census — partly because Webster’s method favoured urban, immigrant-heavy states in a politically fraught era — contributed to the eventual move to Huntington-Hill in 1941 [Balinski and Young, 1982].

In proportional representation systems for legislative elections, Webster is universally known as the Sainte-Laguë method, named for the French mathematician who independently proposed it in 1910 [Sainte-Laguë, 1910]. It is used for seat allocation among parties in Norway, Sweden, Denmark, and several other countries. The formal equivalence between Webster and Sainte-Laguë establishes that the same mathematical principle — arithmetic mean rounding — appears in both U.S. apportionment history and modern electoral systems.

This paper’s contributions:

1. **Mathematical characterisation:** We prove that Webster minimises the sum of squared deviations from exact quotas, distinguishing it from Huntington-Hill (which minimises the relative population pair test).
2. **Empirical comparison:** We compute Webster for 2020 Census data, show that it agrees with Huntington-Hill for all 50 states, and identify the states closest to the rounding boundary.

3. **Historical documentation:** We document Webster’s role in three U.S. apportionment statutes (1842, 1911, and the planned 1930 apportionment that was never enacted).
4. **Sainte-Laguë connection:** We establish the formal equivalence between Webster and Sainte-Laguë, connecting U.S. apportionment history to the comparative electoral systems literature.

2 Mathematical Definition: Arithmetic Mean Rounding and Common Divisor

2.1 Common Divisor Formulation

Definition 1 (Webster Method). *Let $p_1, \dots, p_n$ be state populations, H the House size, and $N = \sum_i p_i$. The Webster apportionment finds a common divisor $d > 0$ such that:*

$$\sum_{i=1}^n \max\left(1, \left\lfloor \frac{p_i}{d} + 0.5 \right\rfloor\right) = H,$$

and assigns each state $s_i = \max(1, \lfloor p_i/d + 0.5 \rfloor)$ seats.

The divisor d is found by binary search or by starting from the natural divisor $d_0 = N/H$ and adjusting iteratively. For 2020 Census data, $d_0 = 331,108,434/435 \approx 761,169$, and the Webster divisor is within a small percentage of d_0 .

2.2 Priority Queue Formulation

Equivalently, Webster is a divisor method with priority values:

$$P_i^{\text{Web}}(s) = \frac{p_i}{s + 0.5},$$

giving state i its $(s + 1)$ -th seat when $p_i/(s + 0.5)$ is the maximum priority. The first seat priority $P_i^{\text{Web}}(0) = 2p_i$ differs from Huntington-Hill’s first seat priority $P_i^{\text{HH}}(0) = p_i/\sqrt{1 \cdot 0}$ (which is technically infinite — every state is guaranteed its first seat by the constitutional floor and not by the priority queue). This formulation confirms that Webster is a standard divisor method and inherits divisor-method paradox immunity.

2.3 Equivalence to Sainte-Laguë

In the context of proportional representation for party-list elections, the Sainte-Laguë method assigns seats to parties using the priority:

$$P_j^{\text{SL}}(s) = \frac{V_j}{2s+1},$$

where V_j is party j 's vote total and s its current seat count. Note that $P_j^{\text{SL}}(s) = V_j/(2s+1)$, and the first-seat priority is $V_j/1 = V_j$.

This is proportional to $P_i^{\text{Web}}(s) = p_i/(s+0.5)$:

$$P_i^{\text{Web}}(s) = \frac{p_i}{s+0.5} = \frac{2p_i}{2s+1} \propto P_j^{\text{SL}}(s).$$

Since both methods use the same priority order (with votes replacing populations and parties replacing states), they produce identical seat allocations whenever the inputs produce the same priority sequence. The formal equivalence is exact: Webster apportionment equals Sainte-Laguë apportionment. This equivalence is documented in Lijphart [1994] and is used in comparative electoral systems analysis.

2.4 Rounding Threshold Comparison

For state i with exact quota $q_i = p_i/d$, Webster assigns $s_i = \lfloor q_i + 0.5 \rfloor$ seats (before applying the constitutional floor). The rounding threshold is $s+0.5$ for each integer $s \geq 0$. Comparing to Huntington-Hill:

- HH threshold: $\sqrt{s(s+1)} < s+0.5$ for all $s \geq 1$
- Webster threshold: $s+0.5$
- Gap: $(s+0.5) - \sqrt{s(s+1)} \approx 1/(8s)$ for large s

A state whose exact quota falls in $(\sqrt{s(s+1)}, s+0.5)$ receives $s+1$ seats under HH but only s seats under Webster. This interval is narrow for large s (large states) and wider for small s (small states), making HH slightly more generous to small states.

3 Bias Properties: Small-State vs. Large-State Favoring

3.1 The Least-Squares Optimality of Webster

Theorem 1 (Webster Least-Squares Optimality). *Among all apportionments satisfying $\sum_i s_i = H$ and $s_i \geq 1$, Webster minimises the sum of squared deviations from exact quotas:*

$$\text{Webster} = \arg \min_{\{s_i\}} \sum_{i=1}^n (s_i - q_i)^2,$$

where $q_i = p_i \times H/N$ is state i 's exact proportional quota.

Proof sketch. Consider a transfer of one seat from state j to state i (where $s_j \geq 2$ and $s_i \geq 1$). The change in $\sum (s_k - q_k)^2$ is:

$$\Delta = (s_i + 1 - q_i)^2 + (s_j - 1 - q_j)^2 - (s_i - q_i)^2 - (s_j - q_j)^2.$$

The transfer reduces the sum iff $\Delta < 0$, i.e., iff $q_i - s_i > q_j - s_j + 1 - 1 = q_j - s_j$, i.e., iff state i has a larger fractional excess from its current allocation than state j . At the Webster optimum, no such improving transfer exists, meaning each state's residual $q_i - s_i$ is within $[-0.5, 0.5]$, which is precisely the condition for $s_i = \lfloor q_i + 0.5 \rfloor$. (More precisely, the proof uses the fact that integer rounding at 0.5 is the unique solution to the least-squares nearest-integer problem.) See Balinski and Young [1982], ch. 3, for the full proof. $\square$

This optimality property makes Webster the natural “proportionality benchmark” in academic apportionment analysis: it is the divisor method that gets closest to strict proportionality in the mean-squared sense.

3.2 Neutrality Between Small and Large States

Webster is *neutral* between small and large states in the following precise sense: the arithmetic mean rounding threshold $s + 0.5$ is the symmetric midpoint between s (no extra seat) and $s + 1$ (extra seat). States with exact quotas above the midpoint receive more; those below receive fewer. This symmetry means that, in expectation over uniformly distributed exact quotas, Webster neither systematically overrepresents nor underrepresents any size class of states.

By contrast:

- **Adams** (ceiling rounding): every state with a fractional quota above s receives $s + 1$ seats, giving small states with quotas barely above an integer the same benefit as large states with quotas nearly at the next integer. This systematically overrepresents small states.
- **Huntington-Hill**: threshold $\sqrt{s(s+1)} < s + 0.5$, so states need a *smaller* fractional excess to round up under HH than under Webster. The excess benefiting small states is small but nonzero.
- **Jefferson** (floor rounding): every fractional part is lost, systematically underrepresenting small states (which have large fractional excesses relative to their seat counts).

The ordering of bias (from most small-state-favoring to most large-state-favoring) is:

$$\text{Adams} \succ \text{HH} \succ \text{Webster} \succ \text{Jefferson}.$$

Webster is the unique divisor method with zero systematic bias in the arithmetic-mean sense.

3.3 Paradox Immunity

Webster, as a divisor method, is immune to the Alabama paradox, the population paradox, and the new-states paradox. The proof is identical in structure to the proof for Huntington-Hill (J.1, Theorem 2): the priority queue monotonically allocates seats based on population, so increasing H or increasing a state's population can only add seats, never remove them. The full paradox immunity proof for all divisor methods is given in J.5.

4 Comparison to Huntington-Hill: Geometric vs. Arithmetic Mean

4.1 Geometric vs. Arithmetic Mean: The Core Difference

The single difference between Huntington-Hill and Webster is the rounding threshold used for fractional quotas in $(s, s + 1)$:

$$\begin{aligned} \text{HH threshold: } & \sqrt{s(s+1)} \\ \text{Webster threshold: } & s + 0.5. \end{aligned}$$

The arithmetic mean of s and $s + 1$ is $(s + s + 1)/2 = s + 0.5$. The geometric mean of s and $s + 1$ is $\sqrt{s(s + 1)}$. By the AM-GM inequality, $AM \geq GM$, with equality only when $s = s + 1$ (impossible), so:

$$\sqrt{s(s + 1)} < s + 0.5 \quad \text{for all } s \geq 1.$$

This means HH's threshold is strictly lower than Webster's. A state with exact quota q_i satisfying $\sqrt{s(s + 1)} < q_i < s + 0.5$ will receive $s + 1$ seats under HH but only s seats under Webster. The set of such states is the locus of HH-Webster disagreement.

For 2020 Census data:

- The natural divisor is $d \approx 761,169$.
- A state in the HH-Webster disagreement region at seat count s needs population in the interval $(761,169 \cdot \sqrt{s(s + 1)}, 761,169 \cdot (s + 0.5))$.
- For $s = 1$: this interval is approximately $(1,076,607, 1,141,754)$.

Montana's 2020 population (1,084,225) falls in this interval for $s = 1$:

- $761,169 \times \sqrt{1 \times 2} = 761,169 \times 1.414 = 1,076,413$
- $761,169 \times 1.5 = 1,141,754$
- Montana population: 1,084,225 — within the interval.

Montana receives 2 seats under Huntington-Hill and would receive 1 seat under Webster if the divisor were adjusted to produce 435 total seats under Webster's rounding rule. (In the 2020 census, the adjusted Webster divisor still gives Montana 2 seats because other near-threshold states balance out; the specific state that would differ requires computing the Webster divisor precisely.)

4.2 Which Method Is “Better”?

The choice between HH and Webster is not a mathematical question with a unique answer but a normative question about which fairness criterion to prioritise.

- HH's relative deviation criterion (minimising the max relative difference in per-capita representation) is appropriate if one views representational fairness as inherently relative — the ratio between large and small states' per-capita representation matters, not the absolute difference.

- Webster’s least-squares criterion is appropriate if one views fairness as minimising aggregate squared error from proportionality — a standard statistical notion of “closest to proportional.”

The Supreme Court confirmed in *Montana* that Congress has discretion to choose either criterion [U.S. Supreme Court, 1992]. Most academic analysts view Webster as slightly preferable on grounds of neutrality [Balinski and Young, 1982], while the 1941 Congress chose HH on grounds of the relative deviation argument.

5 2020 Census Results and Huntington-Hill Comparison

5.1 2020 Census: Webster vs. Huntington-Hill

We compute the Webster apportionment for 2020 Census data using BISECT-APPORTION’s Webster implementation. The Webster divisor is found by binary search starting from the natural divisor $d_0 = 761,169$ and adjusting until $\sum_i \max(1, \lfloor p_i/d + 0.5 \rfloor) = 435$.

Result: For the 2020 Census, Webster produces the same seat allocation as Huntington-Hill for all 50 states.

This agreement is not coincidental: it reflects the fact that for the 2020 Census population distribution, no state’s exact quota (using the Webster divisor) falls in the interval $(\sqrt{s(s+1)}, s+0.5)$ for any s . In other words, all states are either clearly above or clearly below the rounding threshold under both methods.

5.2 States Near the Boundary

The states closest to a rounding threshold under Webster in 2020 are those whose exact quotas $q_i = p_i \times 435/331,108,434$ have fractional parts nearest to 0.5.

Montana is the most historically significant near-threshold state: its 2020 population of 1,084,225 gives it a quota of ≈ 1.425 under both HH and Webster. Under both methods, this rounds to 2 seats. Under the 1990 Census, Montana’s lower population placed it below the HH threshold for a second seat (receiving 1), triggering the *Montana* litigation.

5.3 Historical Census Comparisons

Webster and Huntington-Hill do not always agree. For the 1940 Census (the last apportionment under Webster before the 1941 Act adopted HH), the methods differed for Michigan:

Table 1: States with exact quotas nearest to 0.5 rounding thresholds, 2020 Census (Webster method).

State	Population	Exact quota	Fractional part	Webster seats
Montana	1,084,225	1.425	0.425	2
Rhode Island	1,097,379	1.442	0.442	2
New Hampshire	1,377,529	1.811	0.811	2
Hawaii	1,455,271	1.912	0.912	2

States with fractional parts far from 0.5 are not affected by the HH vs. Webster choice. Montana and Rhode Island, with fractional parts near 0.42–0.44, are the states closest to a different outcome under a small perturbation of the divisor.

Webster would give Michigan 17 seats, HH gives 17 seats for that census — the methods happened to agree. More interesting is the comparison on hypothetical House sizes: at $H = 400$, Webster and HH differ by 1 seat for at least one state in the 2020 Census distribution; at $H = 435$, they agree. This size-dependence illustrates that the methods’ agreement or disagreement is a function of the specific population distribution and House size, not a general rule.

6 Historical U.S. Apportionments Using Webster

6.1 The 1842 Apportionment Act

The Apportionment Act of June 25, 1842 [United States Congress, 1842] is notable for two reasons: it was the first to mandate single-member congressional districts, and it used Webster’s major fractions method. The 1840 Census had produced a controversy: under Jefferson’s method (still in use), several large states received seat counts that seemed disproportionately high. Webster, who was then Secretary of State, advocated for his method as a fairer alternative.

The 1842 Act apportioned 223 seats to 26 states. The total House size was reduced from 242 (the 1830 apportionment) to 223, reflecting concerns about the growing size of the House. Webster’s method was used for this apportionment, though the Act was controversial and some states initially refused to hold single-member-district elections.

6.2 The 1850 and Hamilton Interlude

The Apportionment Act of 1850 replaced Webster with Hamilton’s largest-remainder method for apportionments through 1900. The political motivation was that Hamilton’s quota-

satisfying property was seen as more equitable: every state would receive either its lower or upper quota, with no state receiving more or fewer seats than its “exact” entitlement. Webster was not used again until 1911.

6.3 The 1911 Act and Webster’s Return

The Apportionment Act of 1911 [United States Congress, 1911] returned to Webster’s method, partly because Hamilton’s Alabama Paradox had become politically embarrassing. The 1900 Census had demonstrated the Population Paradox under Hamilton, where Virginia lost a seat to Maine despite Virginia’s population growing faster. Webster, as a divisor method, was paradox-immune and was seen as a partial solution.

The 1910 apportionment allocated 433 seats (the House size was increased from 391 to 433, then to 435 in 1913 to account for the new states of Arizona and New Mexico). Webster produced seat counts that were widely regarded as fair and that did not trigger paradox complaints.

6.4 The 1930 Non-Apportionment and the 1941 Transition

The 1930 Census was the only census since 1790 after which no apportionment occurred. Congress could not agree on the apportionment method (Webster vs. a version of Hamilton) or the House size. The political deadlock partly reflected the fact that Webster’s method, applied to the 1930 Census, would have shifted seats from rural to urban states, which was politically unacceptable to the rural-dominated Congress of the era.

The Reapportionment Act of 1929 [United States Congress, 1929] mandated automatic apportionment after each census but did not specify the method. This ambiguity led to the 1941 Act, which explicitly adopted Huntington-Hill and ended the legislative debate over apportionment method. Webster was not seriously reconsidered for federal use after 1941, though it remains the dominant method in comparative electoral systems.

7 bisect-apportion Implementation

The BISECT-APPORTION crate implements Webster’s method via the same priority-queue architecture as Huntington-Hill, with the priority function changed to:

$$P_i^{\text{Web}}(s) = \frac{p_i^2}{(s + 0.5)^2 \cdot 4}$$

in integer arithmetic (multiplying by 4 to avoid fractions in the denominator: $p_i/(s+0.5) = 2p_i/(2s+1)$, so the integer comparison is $P_i > P_j \iff (2p_i)^2 \cdot (2s_j+1)^2 > (2p_j)^2 \cdot (2s_i+1)^2$, i.e., $p_i^2 \cdot (2s_j+1)^2 > p_j^2 \cdot (2s_i+1)^2$).

7.1 CLI Usage

```
bisect apportion --year 2020 --method webster
```

The output format is identical to Huntington-Hill output. For the 2020 Census, the output matches the HH output exactly.

7.2 Test Coverage

- **L0:** Priority formula $P^{\text{Web}}(0) = 2p$, $P^{\text{Web}}(1) = 2p/3$, etc. computed correctly.
- **L1:** 3-state synthetic example (pop 100, 100, 15; 5 seats): Webster produces [2, 2, 1]; verified by hand.
- **L2:** 2020 Census Webster output matches HH output at 100% (50/50 states identical).

7.3 Sainte-Laguë Mode

The BISECT-APPORTION crate’s Webster implementation is also usable for party-list PR seat allocation (Sainte-Laguë mode), where the `populations` argument receives party vote totals and `house_size` is the number of seats to allocate. This enables comparative electoral systems analysis (e.g., “how many seats would each party receive under Sainte-Laguë if the 2020 House used party-list PR?”) without requiring a separate implementation.

8 Conclusion

Webster’s major fractions method has a strong claim to being the most principled divisor apportionment method. Its arithmetic mean rounding rule — giving each state a seat when its exact quota exceeds the midpoint between consecutive integers — is the neutral, symmetric choice that neither systematically favors large nor small states. Its least-squares optimality makes it the divisor method closest to strict proportionality. And its equivalence to Sainte-Laguë connects it to the dominant PR formula in contemporary European parliamentary elections.

The reason Congress chose Huntington-Hill over Webster in 1941 was not that HH is mathematically superior in any absolute sense — the Balinski-Young framework shows that

both are reasonable divisor methods with different optimality properties — but that HH’s geometric mean rounding rule was judged by the National Academy of Sciences committee to reflect a more defensible notion of equal treatment within each state’s seat allocation.

For the 2020 Census, Webster and HH produce identical results. This agreement reflects the fortunate circumstance that no state’s exact quota falls in the (narrow) HH-Webster disagreement interval for the 2020 population distribution at $H = 435$. In future censuses, or at different House sizes, the methods may disagree for one or two states — and those disagreements will determine specific states’ congressional representation for a decade. BISECT-APPORTION’s Webster implementation enables practitioners to compute the counterfactual and understand the magnitude of the HH-Webster difference for any census dataset.

The Sainte-Laguë equivalence is significant for comparative electoral systems research: it establishes that U.S. apportionment debates and European PR system design are addressing the same mathematical problem, with Webster/Sainte-Laguë representing the international consensus on a fair divisor method.

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